



FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES

ICT 502 : DATABASE ENGINEERING

TITLE:

DIBO KITCHEN



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PREPARED FOR : SIR MUHAMMAD HAMIZ BIN MOHD RADZI

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1.0 INTRODUCTION

1.1 Company Background



In the beginning of 2024, a cute and lovely lady with full passion in cooking named Adibah started her journey by opening DiBo Kitchen. This restaurant is based in the lively town of Kajang, Selangor. She opened this restaurant because she wants to broaden her passion in cooking and share her unique cooking taste with others. With the marvelous taste in food, DiBo's Kitchen quickly became a popular destination from people all over the country.

DiBo Kitchen, which is perfectly named by the owner, it serves a mouth-watering blend of Thai and western cuisines. It also offers a wide variety of food that suits with every Malaysian's taste, including their hot selling western foods like Carbonara Pasta and burgers as well as aromatic Thai curries that originally same as in Thailand.

In terms of location, DiBo Kitchen, which is centrally located in Kajang, offers a cozy and welcoming dining area where customers may enjoy every single meals there. The main objective was that the owner opened this restaurant because she wanted the eating experience to be exactly the same as dining at a friend's house, and this objective was successfully achieved by its friendly atmosphere and accommodating staff members.

Adibah, with her dedicated staff at DiBo Kitchen, are fully committed to deliver appetizing meals prepared with the finest ingredients and authentic tastes. To be more specific, the tastes combines the richness of western classics and freshness of Thai spices, in every meals they served is made with fully love and attention.

2.0 CASE STUDY

2.1 Problem Statement

Dibo Kitchen is currently facing several challenges with its reservation and data management system, which has affected the efficient operation of the restaurant. The existing system lacks a structured approach for managing reservations, resulting in conflicts and inefficiencies during peak hours. Additionally, obtaining accurate information about staff availability is cumbersome, leading to difficulties in planning and resource allocation. Moreover, the reliance on paper-based records for customer data has resulted in poor records management, making it challenging to track and analyze customer behavior effectively.

These challenges highlight the urgent need for a comprehensive reservation and data management solution at Dibo Kitchen. The new system must address the following key issues:

1. **Inefficient Reservation System:** The current reservation system at Dibo Kitchen lacks the capability for customers to book specific time slots, leading to conflicts and time clashes, particularly during peak hours. This inefficiency not only affects customer experience but also impacts operational efficiency.
2. **Lack of Staff Information:** Dibo Kitchen faces difficulties in obtaining accurate information about staff schedules and availability, making it challenging to plan and allocate resources effectively. Implementing a systematic routine for updating staff schedules and availability is essential to ensure the integrity of the database and improve operational efficiency.
3. **Poor Records Management:** The reliance on paper-based records for customer data results in inadequate records management, making it difficult to track and analyze customer behavior effectively. This lack of comprehensive records management poses risks such as data loss, inaccuracies, compliance shortcomings, and operational inefficiencies.
 - i. **Data Loss and Inaccuracy:** When records aren't managed well, there's a higher chance of accidentally deleting or changing vital details, which can cause confusion and errors throughout the organization. This could mean making decisions based on incorrect information or even costly mistakes that could have been prevented. So, it's essential to have reliable systems in place to keep records accurate and up-to-date, reducing the chances of these problems occurring.

- ii. Compliance shortcomings: The organization is not meeting the necessary legal or regulatory requirements, potentially exposing it to legal or reputational risks.
- iii. Operational inefficiencies: The organization's processes are not running as smoothly or efficiently as they could be, leading to wasted resources or reduced productivity.

To address these challenges, Dibo Kitchen requires a strong reservation and data management system that streamlines reservation processes, provides accurate information about staff schedules and availability, and improves records management practices. By implementing such a solution, Dibo Kitchen aims to enhance operational efficiency, improve customer experience, and optimize resource utilization.

2.2 Objectives

1. Analyze

To analyze the restaurant reservation process and staff management needs which are the information required for reservations, such as customer details and reservation's packages, as well as staff-in-charge's information.

2. Design

To design a database system that reduces data redundancy and data inconsistent involving the reservation records, increase productivity of the package selection process for customers.

3. Develop

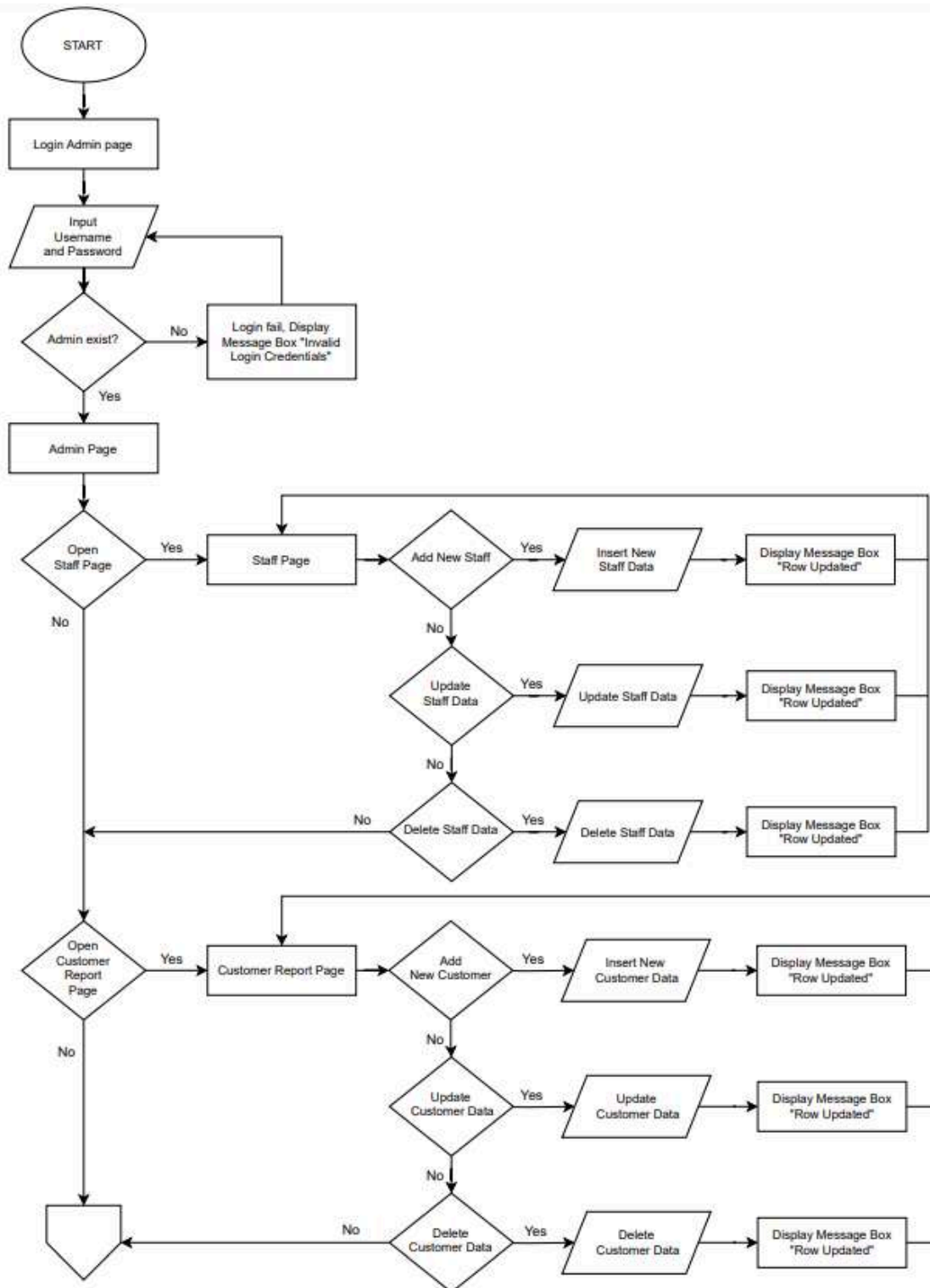
To develop a database system that allows the customer to make a reservation online and reserve specific packages to order and to help staff to manage or handle the activities that are involved in the reservation process in an organized and cost-effective way.

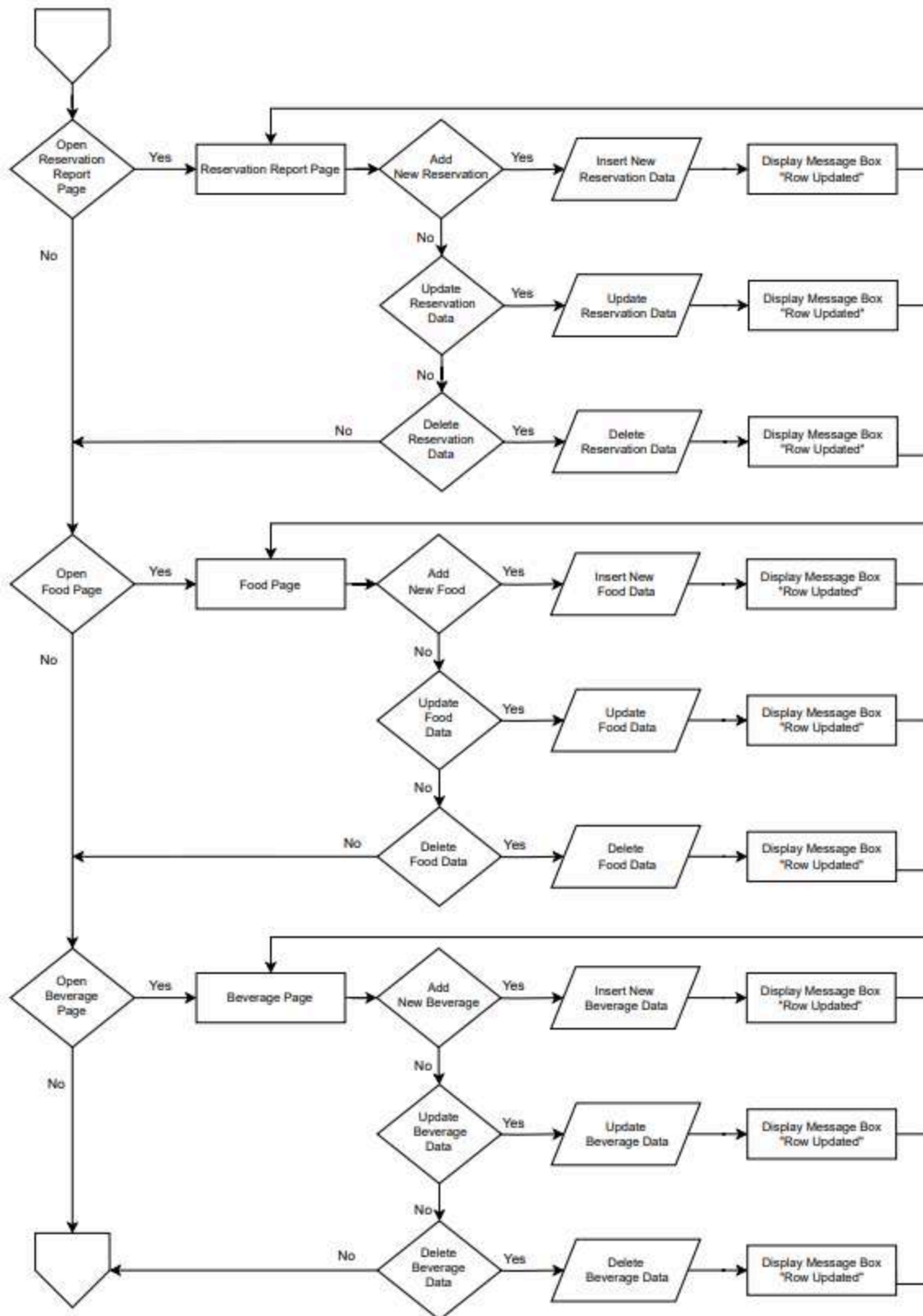
4. Test

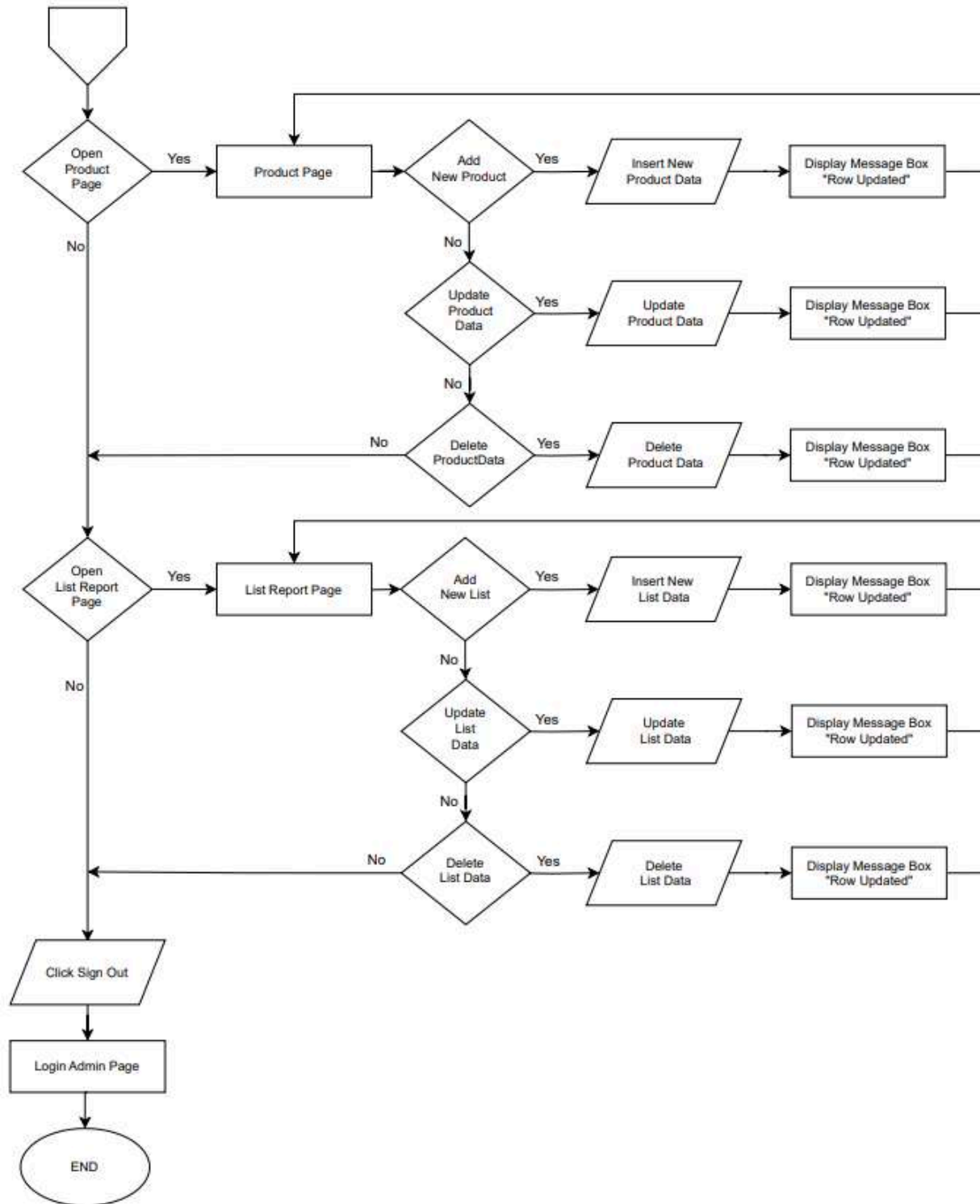
To test the functionality of this database system of daily activities in a restaurant.

3.0 SYSTEM DESIGN

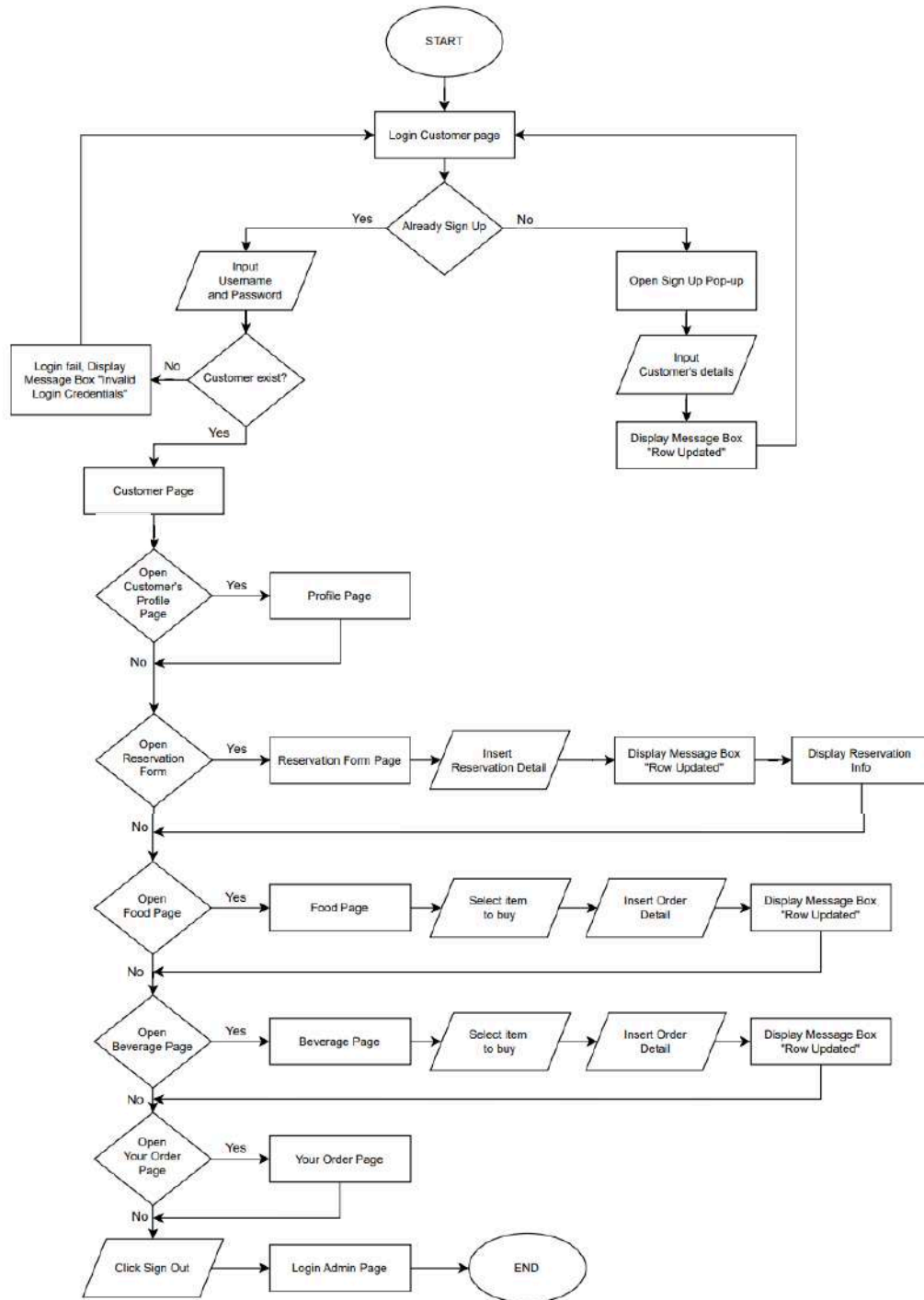
3.1.1 Flow Chart of System (Admin)







3.1.2 Flow Chart of Svstem (Customer)



3.2 10 SQL Queries

- 1) Find the full names of the staff who are managers.

```
1 select staff_fname||' '||staff_lname "STAFF NAME"
2 from staff
3 where manager_id is null;
```

Results	Explain	Describe	Saved SQL	History
STAFF NAME				
Ikhmal Arif				
Ahmad Amirul				
2 rows returned in 0.01 seconds Download				

- 2) Display the customer name who makes a reservation in June.

```
1 SELECT cust_fname||' '||cust_lname "CUSTOMER", res_date
2 FROM reservation r
3 join customer c
4 on c.cust_id=r.cust_id
5 WHERE TO_CHAR(res_date, 'MM') = 06;
6
```

Results	Explain	Describe	Saved SQL	History
CUSTOMER		RES_DATE		
Muhammad Abdullah		06/13/2024		
Muhammad Abdullah		06/27/2024		
Muhammad Abdullah		06/30/2024		
Muhammad Abdullah		06/28/2024		
4 rows returned in 0.01 seconds Download				

- 3) Display the product ID and product name for products that contain 'Thai' in the product description.

```
1 select prod_id, prod_name
2 from product
3 where prod_desc like '%Thai%';
```

Results

Explain

Describe

Saved SQL

History

PROD_ID	PROD_NAME
3	Massaman Curry
8	Green Thai Tea

2 rows returned in 0.00 seconds [Download](#)

- 4) Display the number of reservations made by each staff member:

```

1 select s.staff_id, s.staff_fname, s.staff_lname, count(r.res_id) as "Number of Reservations"
2 from staff s
3 join reservation r
4 on s.staff_id=r.staff_id
5 group by s.staff_id, s.staff_fname, s.staff_lname;

```

STAFF_ID	STAFF_FNAME	STAFF_LNAME	Number of Reservations
4	Ahmad	Amirul	1
5	Ikhmal	Arif	1
12	Leona	Kasih	1

3 rows returned in 0.02 seconds [Download](#)

- 5) Display the top 2 products by total quantity sold.

```

1 select prod_id, prod_name, prod_price, prod_type
2 from product
3 where prod_id in (
4     select prod_id
5     from (
6         select prod_id, sum(prod_quantity) as total_quantity
7         from list
8         group by prod_id
9         order by total_quantity desc
10        )
11 where rownum <= 2
12 );

```

PROD_ID	PROD_NAME	PROD_PRICE	PROD_TYPE
1	Nasi Goreng	10	FOOD
5	Iced Lemonade	4	BEVERAGE

- 6) Display product ID, product name with highest price.

```

1 select prod_id, prod_name
2 from product
3 where prod_price = (select max(prod_price)
4                    from product);

```

PROD_ID	PROD_NAME
11	Spaghetti Carbonara
4	Buffalo Chicken

2 rows returned in 0.01 seconds [Download](#)

- 7) Display customer ID, customer first name and the percentage of total orders placed by each customer

```

1 select c.cust_id, c.cust_fname, count(list_id)*100/ (select count(*) from list) "Percentage of Total Orders"
2 from customer c
3 join reservation r
4 on c.cust_id=r.cust_id
5 join list l
6 on r.res_id=l.res_id
7 group by c.cust_id, c.cust_fname;
8

```

CUST_ID	CUST_FNAME	Percentage of Total Orders
1	Muhammad	25
10	zulaikha	12.5
9	Muhammad	25
3	Siti	37.5

4 rows returned in 0.01 seconds [Download](#)

- 8) Retrieve a list of food and beverage items that have been reserved with the latest reservation date.

```

1 SELECT p.prod_name, r.res_date
2 FROM product p
3 JOIN list l ON p.prod_id = l.prod_id
4 JOIN reservation r ON l.res_id = r.res_id
5 WHERE r.res_date = (SELECT MAX(res_date) FROM reservation);
6

```

PROD_NAME	RES_DATE
Iced Lemonade	07/25/2024
Nasi Goreng	07/25/2024
Nasi Goreng	07/25/2024

3 rows returned in 0.01 seconds [Download](#)

- 9) Find the names of customers who have placed an order with a total value greater than the average total value of all orders.

```

1 SELECT cust_fname || ' ' || cust_lname "Full Name"
2 FROM customer
3 WHERE cust_id IN (SELECT r.cust_id
4 FROM reservation r
5 JOIN list l ON r.res_id = l.res_id
6 JOIN product p ON l.prod_id = p.prod_id
7 GROUP BY r.cust_id
8 HAVING SUM(p.prod_price * l.prod_quantity) > (SELECT AVG(sub.total)
9 FROM (SELECT r.res_id, SUM(p.prod_price * l.prod_quantity) AS total
10 FROM reservation r
11 JOIN list l ON r.res_id = l.res_id
12 JOIN product p ON l.prod_id = p.prod_id
13 GROUP BY r.res_id) sub
14 )
15 );

```

Full Name
zulaikha raof

1 rows returned in 0.03 seconds [Download](#)

10) Display all customer id and customer first name for each reservation.

```
1 select r.staff_id, r.cust_id, c.cust_fname
2 from customer c
3 left outer join reservation r
4 on c.cust_id=r.cust_id;
```

Results	Explain	Describe	Saved SQL	History
STAFF_ID	CUST_ID	CUST_FNAME		
12	1	Muhammad		
10	1	Muhammad		
5	1	Muhammad		
7	1	Muhammad		
7	1	Muhammad		
4	1	Muhammad		
-	-	Nurul		
10	3	Siti		
8	3	Siti		
8	3	Siti		

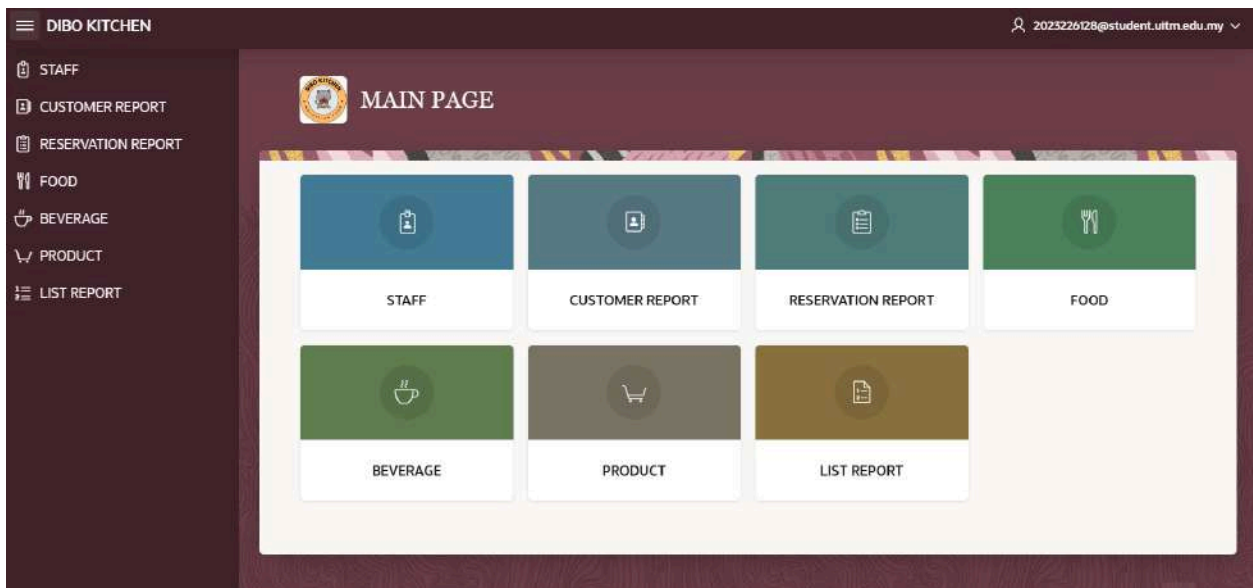
More than 10 rows available. Increase row count to view more rows.

4.0 SYSTEM DEVELOPMENT SAMPLE SCREEN

ADMIN VIEW



LOGIN PAGE



HOME PAGE

DIBO KITCHEN

2023226128@student.uitm.edu.my

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

STAFF

Q

Go

Actions

Create

	Staff ID	Staff First Name	Staff Last Name	Staff Phone Num	Staff Email Address	Manager
	5	Ikhmal	Arif	018-2345678	ikhmal@gmail.com	
	7	Mei	Ling	013-7153927	ling@gmail.com	Ikhmal
	12	Leona	Kasih	016-3638293	leona@gmail.com	Ahmad
	13	Siti	Khairiyah	018-8883379	khairiyah@gmail.com	Ikhmal
	10	Naeiman	Ali	018-3739273	naeiman@gmail.com	Ahmad
	11	Alif	Teega	018-2938372	teega@gmail.com	Ikhmal
	4	Ahmad	Amirul	011-6578935	amirul@gmail.com	
	6	Muhammad	Faiz	019-2637495	muhammad_faiz@gmail.com	Ahmad
	9	Khairul	Anuar	017-6353826	anuarkhairul@gmail.com	Ahmad
	8	Nur	Alia	019_2583628	alia@gmail.com	Ikhmal

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STAFF PAGE

DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

STAFF

Q

Go

Actions

	Staff ID	Staff First Name	Staff Last Name
	5	Ikhmal	Arif
	7	Mei	Ling
	12	Leona	Kasih
	13	Siti	Khairiyah
	10	Naeiman	Ali
	11	Alif	Teega
	4	Ahmad	Amirul
	6	Muhammad	Faiz
	9	Khairul	Anuar
	8	Nur	Alia

STAFF

Staff Fname

Required

Staff Lname

Required

Staff Phone

Required

Staff Email

Required

Manager Id

Cancel

Create

STAFF ADD PAGE

DIBO KITCHEN

- STAFF
- CUSTOMER REPORT
- RESERVATION REPORT
- FOOD
- BEVERAGE
- PRODUCT
- LIST REPORT

STAFF

Staff ID	Staff First Name	Staff Last Name
5	Ikhmal	Arif
7	Mei	Ling
12	Leona	Kasih
13	Siti	Khairiyah
10	Naeiman	Aji
11	Alif	Teega
4	Ahmad	Aminul
6	Muhammad	Falz
9	Khairul	Anuar
8	Nur	Alia

STAFF

Staff Name: Mei

Staff Phone: 013-7153927

Staff Email: ling@gmail.com

Manager Id: Ikhmal Arif

Buttons: Cancel, Delete, Apply Changes

DIBO KITCHEN
 2023226128@student.uitm.edu.my ▼

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

CUSTOMER REPORT

Go Actions ▼

Create

	Cust ID	Cust First Name	Cust Last Name	Cust Phone Num	Cust Email Address	Cust Password
	4	Ahmad	Ibrahim	014-5678901	ahmadibrahim@gmail.com	ibrahimm
	9	Muhammad	Faris	0196707174	muhammad.faris@gmail.com	farist23
	2	Nurul	Huda	019-8765432	nurulhuda@gmail.com	nurul02
	8	nurul	najwa	017-6872588	najwa@gmail.com	najwa
	10	zulaikha	raof	01120817716	zulaikha@gmail.com	zulaikha
	7	Nor	Ain	019-2345678	nor.ain@gmail.com	ainnnn
	1	Muhammad	Abdullah	012-3456789	muhammad.abdullah@gmail.com	abdullah01
	3	Siti	Aishah	017-2345678	siti.aishah@gmail.com	sitiaishah
	5	Mohd	Firdaus	015-4567890	muhd.firdaus@gmail.com	firdaus04
	6	Farah	Nadia	016-7890123	farah.nadia@gmail.com	nadia05

1 - 10

DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

CUSTOMER REPORT

Q

Go

Actions

	Cust ID	Cust First Name	Cust Last Name
☒	4	Ahmad	Ibrahim
☒	9	Muhammad	Faris
☒	2	Nurul	Huda
☒	8	nurul	najwa
☒	10	zulaikha	raof
☒	7	Nor	Ain
☒	1	Muhammad	Abdullah
☒	3	Siti	Aishah
☒	5	Mohd	Firdaus
☒	6	Farah	Nadia

CUSTOMER REPORT

Cust Fname

Required

Cust Lname

Required

Cust Phone

Required

Cust Email

Required

Cust Password

Required

Cancel

Create

CUSTOMER ADD PAGE

DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

CUSTOMER REPORT

Q

Go

Actions

	Cust ID	Cust First Name	Cust Last Name
☒	4	Ahmad	Ibrahim
☒	9	Muhammad	Faris
☒	2	Nurul	Huda
☒	8	nurul	najwa
☒	10	zulaikha	raof
☒	7	Nor	Ain
☒	1	Muhammad	Abdullah
☒	3	Siti	Aishah
☒	5	Mohd	Firdaus
☒	6	Farah	Nadia

CUSTOMER REPORT

Cust Fname

Ahmad

Required

Cust Lname

Ibrahim

Required

Cust Phone

014-5678901

Required

Cust Email

ahmad.ibrahim@gmail.com

Required

Cust Password

ibrahimm

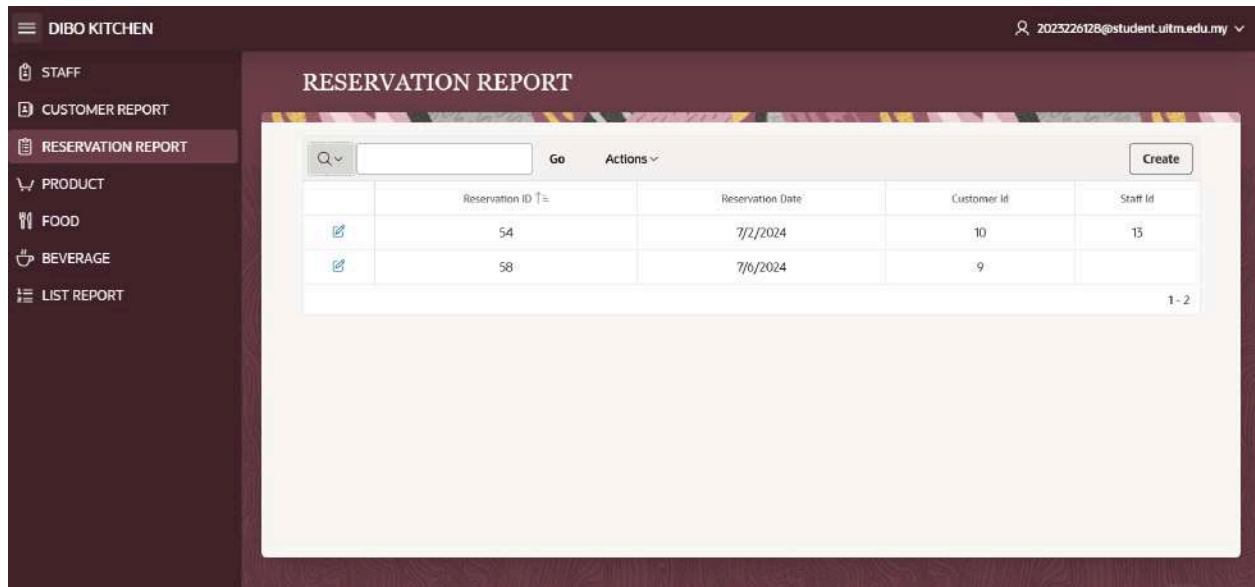
Required

Cancel

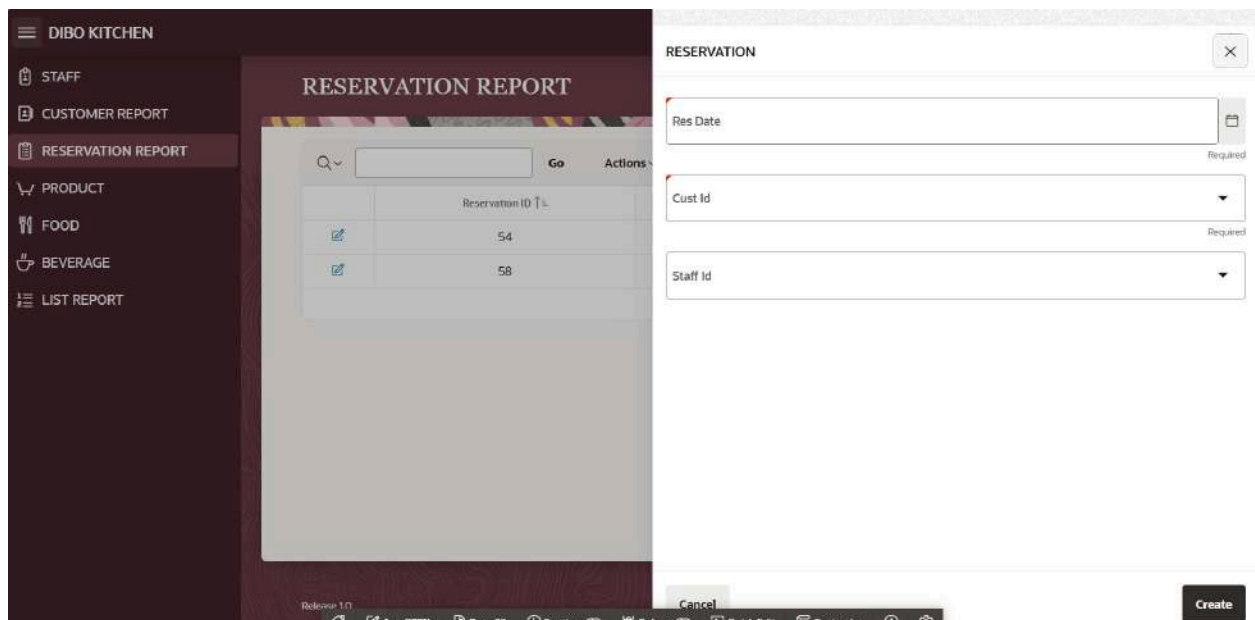
Delete

Apply Changes

CUSTOMER UPDATE PAGE



RESERVATION PAGE



RESERVATION ADD PAGE

DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

PRODUCT

FOOD

BEVERAGE

LIST REPORT

RESERVATION REPORT

Q

Go

Actions

	Reservation ID ↑
	54
	58

RESERVATION

Res Date

7/2/2024

Required

Cust Id

zuleikha

Required

Staff Id

Siti

Cancel

Delete

Apply Changes

RESERVATION UPDATE PAGE

DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

FOOD

Q

Go

Actions

Create

	Product Id	Product Name	Spicy Level	Product Pic
	11	Spaghetti Carbonara	2	
	12	Cheesy Fries	2	

FOOD PAGE

☰

DIBO KITCHEN

📅

STAFF

📅

CUSTOMER REPORT

📅

RESERVATION REPORT

🍴

FOOD

🍹

BEVERAGE

🍽️

PRODUCT

☰

LIST REPORT

🔍

2025226128@student.uitm.edu.my

▼

FOOD \

FOOD2 FORM

FOOD FORM

CHOOSE FOOD

☐ Mee Curry

☐ Spaghetti Carbonara

☐ Pad Thai

☐ Massaman Curry

☐ Thai Chicken Satay

☐ Lasagna

☐ Nasi Goreng

☐ Buffalo Chicken

☐ Grilled Chicken Chop

☐ Cheesy Fries

Spicy Level

☐ 🍶 🍶 🍶 🍶 🍶 0

Required

🏠

📄 App 17776

📄 Page 7

🔌 Session

🔌 Debug

🔌 Quick Edit

🔌 Customize

🔌 1

⚙️

Confirm

≡ DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

2025226128@student.uitm.edu.my

FOOD FORM

CHOOSE FOOD

☐ Mee Curry

☒ Spaghetti Carbonara

☐ Pad Thai

☐ Massaman Curry

☐ Thai Chicken Satay

☐ Lasagna

☐ Nasi Goreng

☐ Buffalo Chicken

☐ Grilled Chicken Chop

☐ Cheesy Fries

Spicy Level

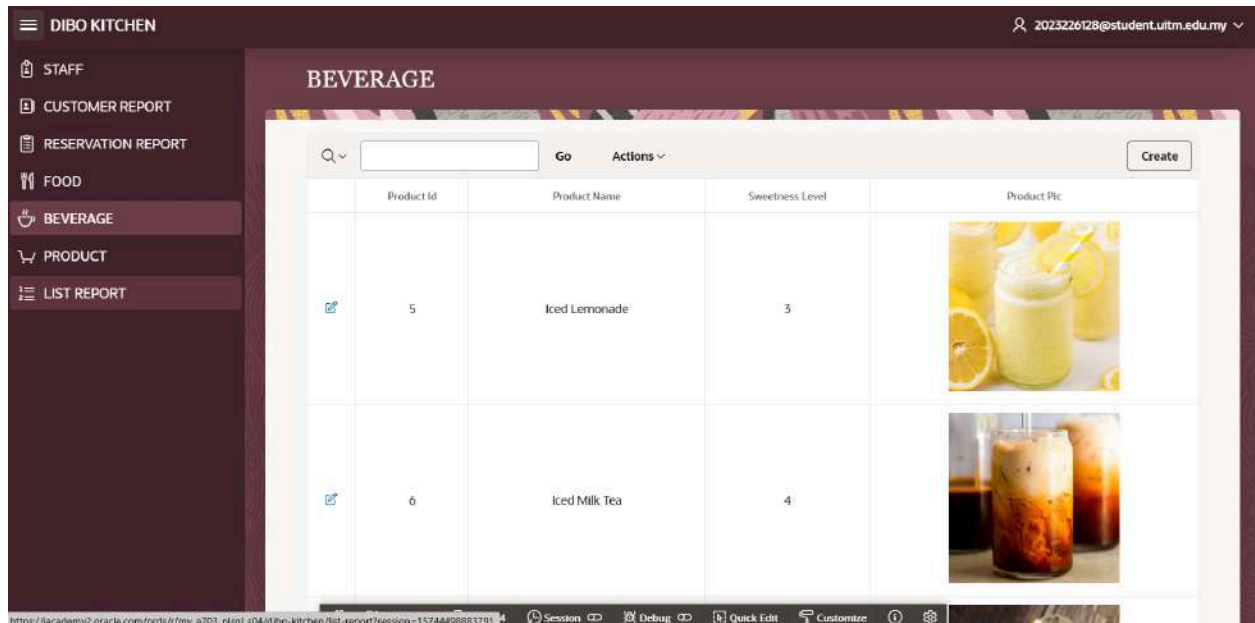
2

Required

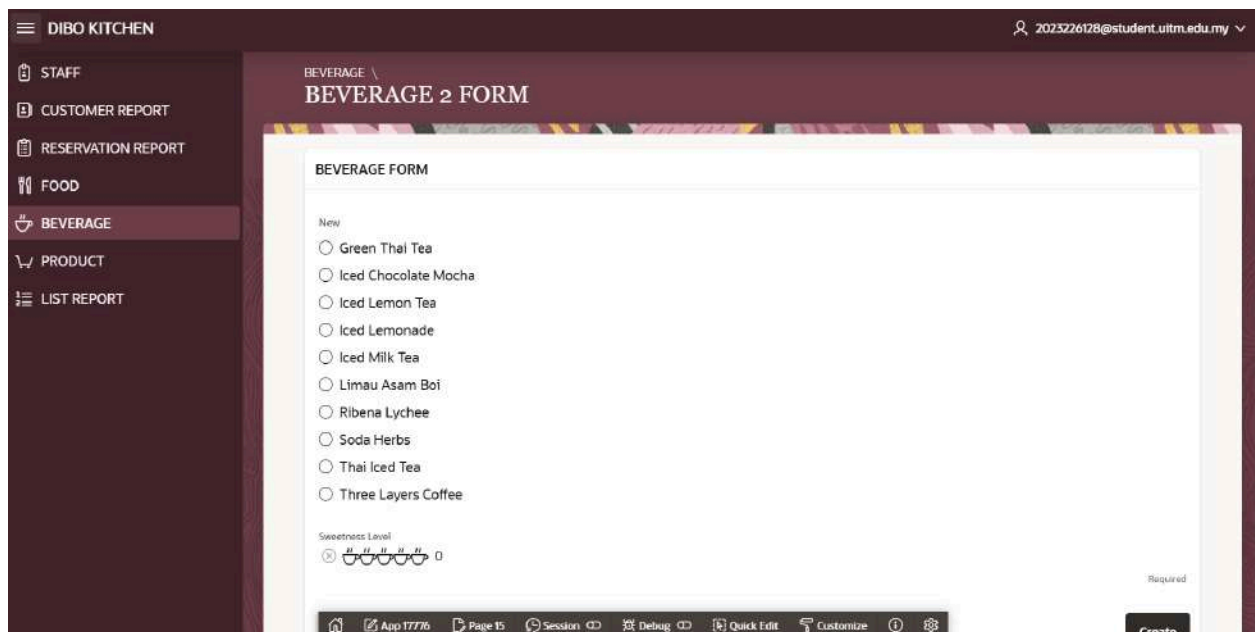
Cancel

Delete

Apply Changes



BEVERAGE PAGE



BEVERAGE ADD PAGE

DIBO KITCHEN

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STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

BEVERAGE

BEVERAGE 2 FORM

BEVERAGE FORM

New

☐ Green Thai Tea
 ☐ Iced Chocolate Mocha
 ☐ Iced Lemon Tea
 ☒ Iced Lemonade
 ☐ Iced Milk Tea
 ☐ Limau Asam Boi
 ☐ Ribena Lychee
 ☐ Soda Herbs
 ☐ Thai Iced Tea
 ☐ Three Layers Coffee

Sweetness Level

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

App 17776

Page 15

Session

Debug

Quick Edit

Customize

Delete

Apply Changes

BEVERAGE UPDATE PAGE

DIBO KITCHEN

2023226128@student.uitm.edu.my

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

PRODUCT

Go

Actions

Create

Product ID	Product Name	Product Description	Product Price	Product Type	Product Picture
10	Iced Chocolate Mocha	A sweeter and more indulgent drink that caters to chocolate lovers!	RM5	BEVERAGE	
9	Mee Curry	A typical preparation of Malaysian/Singaporean-style curry mee	RM7	FOOD	
11	Spaghetti Carbonara	A classic Italian pasta dish combines a silky cheese sauce with pancetta and black pepper	RM17	FOOD	

PRODUCT PAGE

21

DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

PRODUCT

Q

Go

Actions

Product ID	Product Name	Prod
10	Iced Chocolate Mocha	A sweeter and more indulgent
9	Mee Curry	A typical preparation of Ma
11	Spaghetti	A classic Italian pasta dish

PRODUCT

Prod Name

Required

Prod Desc

Prod Price

Required

Prod Pic

Choose File

Prod Type

FOOD

Cancel

Create

PRODUCT ADD PAGE

DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

PRODUCT

Q

Go

Actions

Product ID	Product Name	Prod
10	Iced Chocolate Mocha	A sweeter and more indulgent
9	Mee Curry	A typical preparation of Ma
11	Spaghetti	A classic Italian pasta dish

PRODUCT

Prod Name

Iced Chocolate Mocha

Required

Prod Desc

A sweeter and more indulgent drink that caters to chocolate lovers!

Prod Price

5

Required

Prod Pic

Choose File

Download

Prod Type

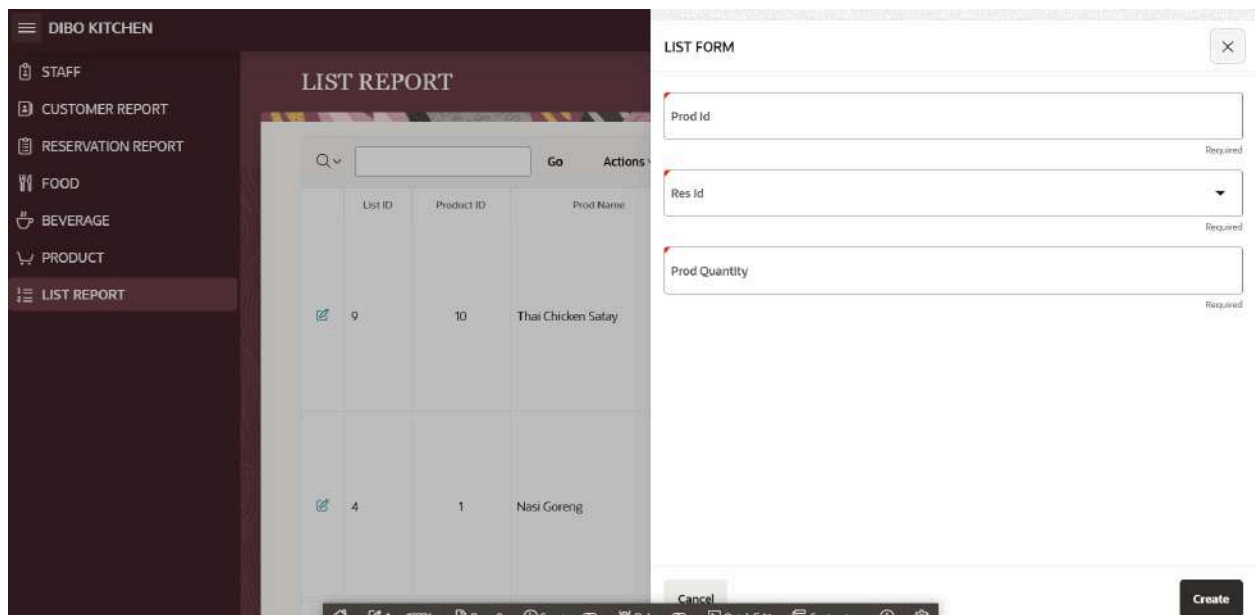
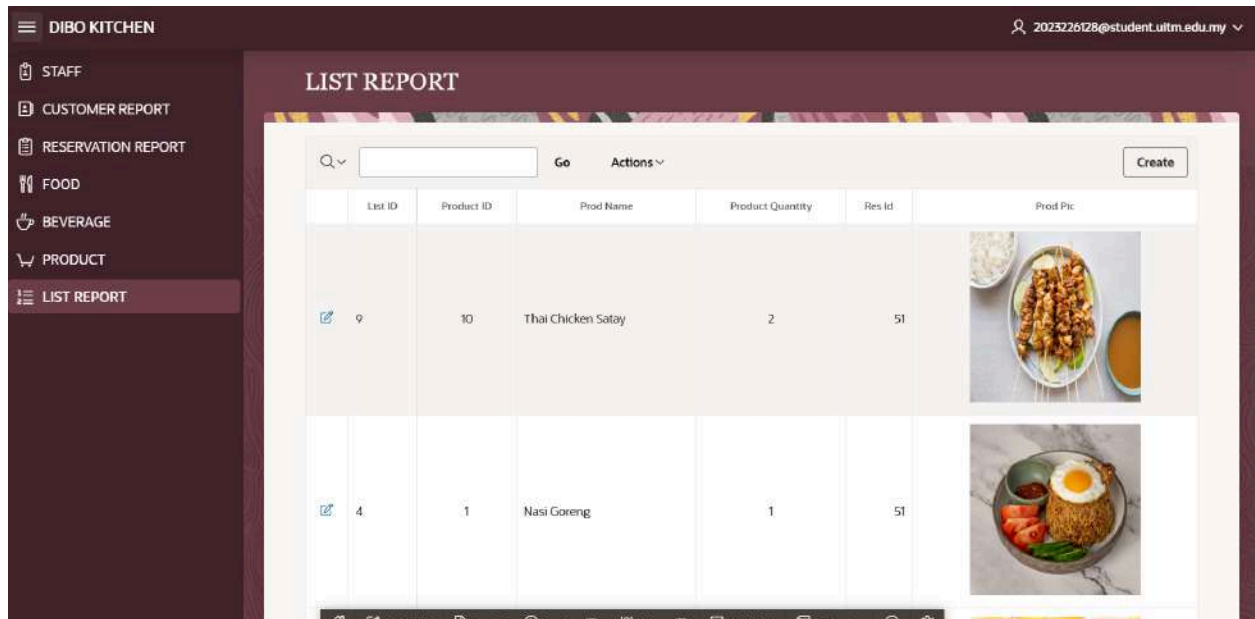
BEVERAGE

Cancel

Delete

Apply Changes

PRODUCT UPDATE PAGE



DIBO KITCHEN

STAFF

CUSTOMER REPORT

RESERVATION REPORT

FOOD

BEVERAGE

PRODUCT

LIST REPORT

LIST REPORT

Q

Go

Actions

	List ID	Product ID	Prod Name
	9	10	Thai Chicken Satay
	4	1	Nasi Goreng

LIST FORM

Prod Id

10

Required

Res Id

51

Required

Prod Quantity

2

Required

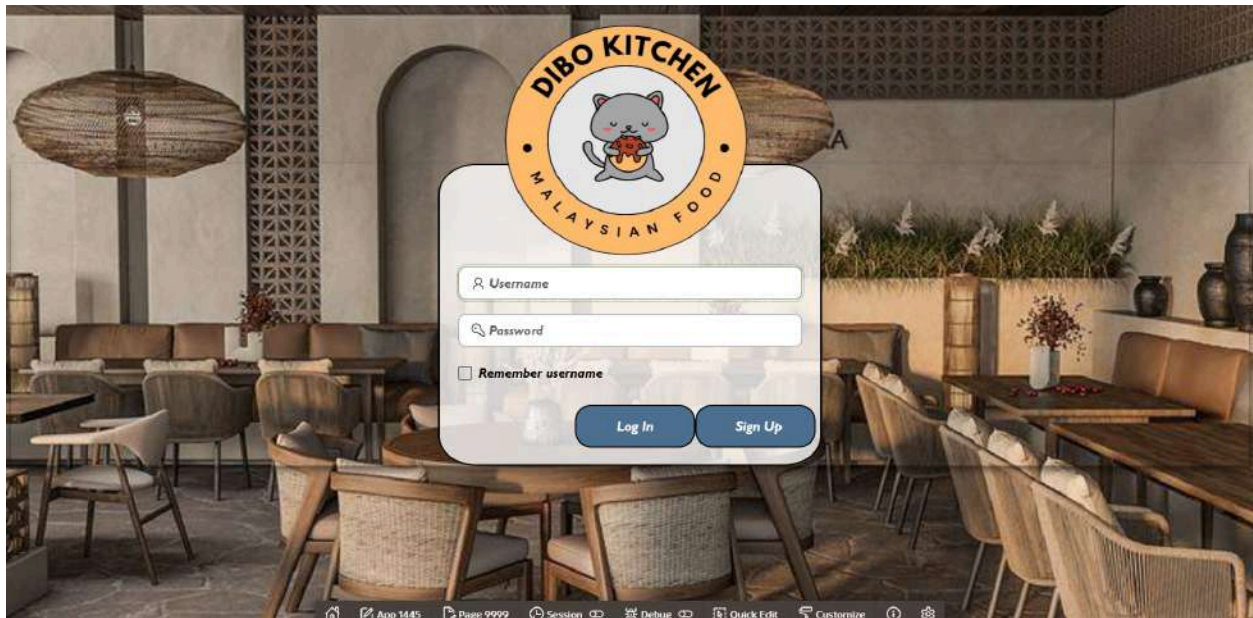
Cancel

Delete

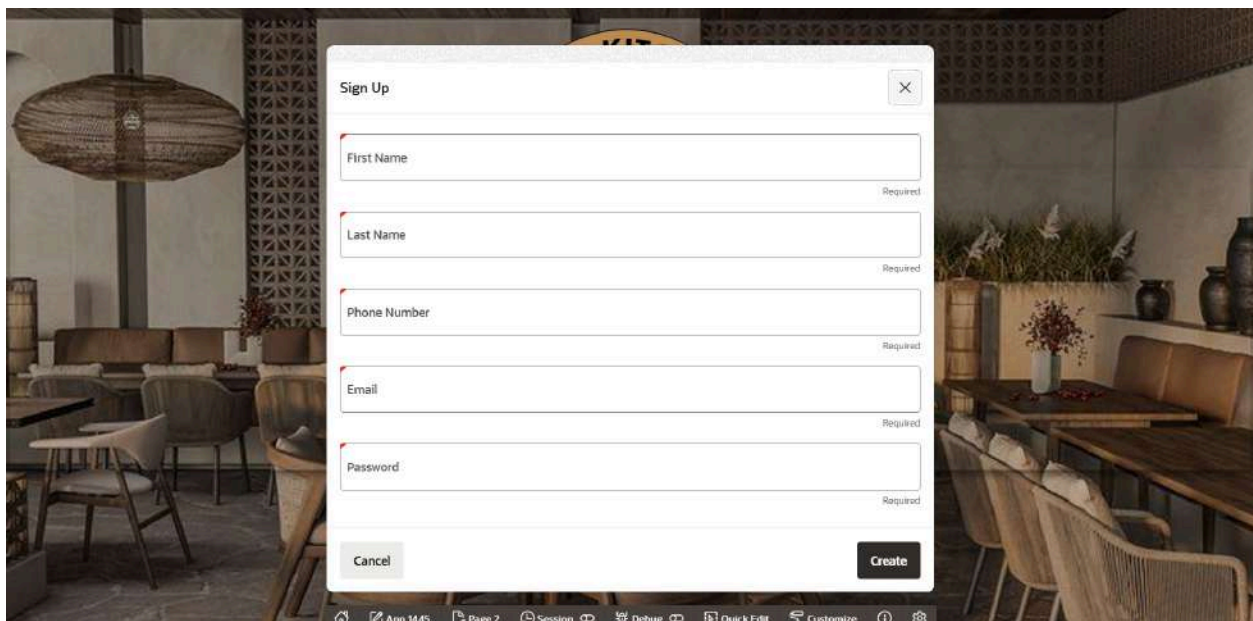
Apply Changes

LIST UPDATE PAGE

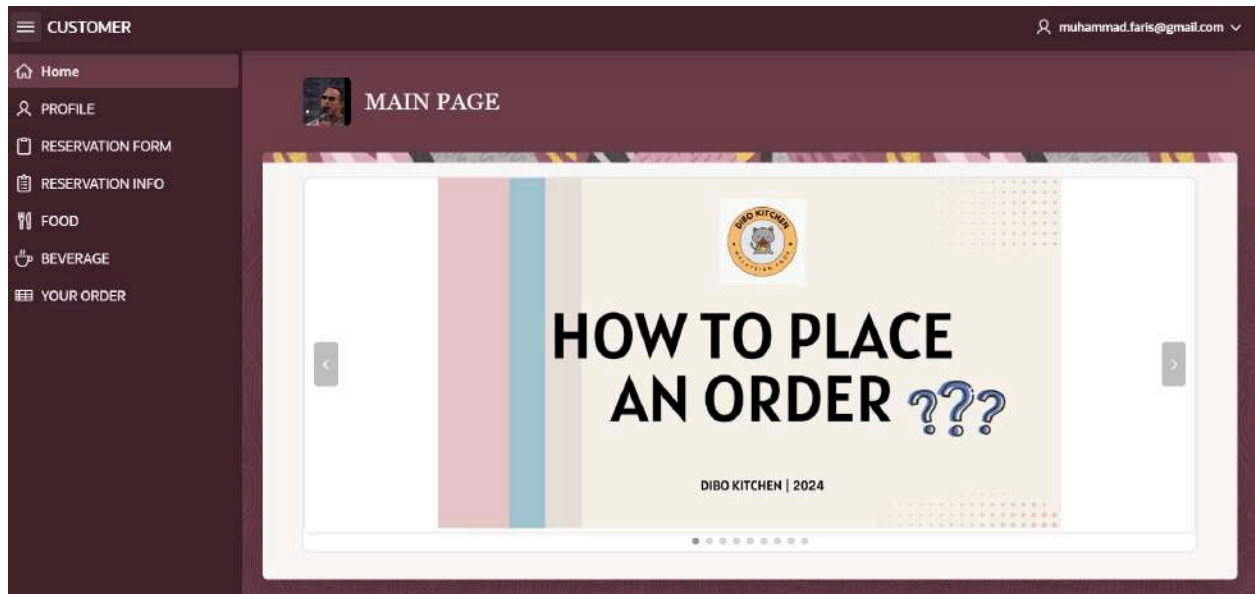
CUSTOMER VIEW



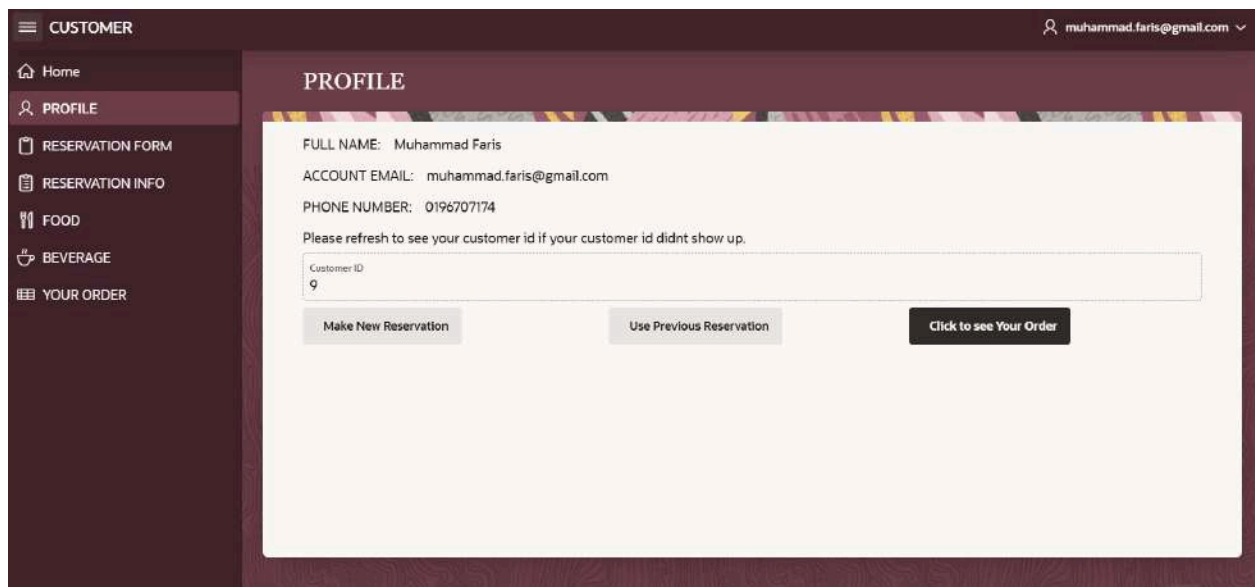
LOGIN PAGE



SIGN UP PAGE



HOME PAGE



PROFILE PAGE

CUSTOMER

Home

PROFILE

RESERVATION FORM

RESERVATION INFO

FOOD

BEVERAGE

YOUR ORDER

siti.alshah@gmail.com

RESERVATION FORM

RESERVATION FORM

Reservation Date

Customer ID

Forgot Customer ID

Use Previous Reservation

Create

RESERVATION FORM PAGE

CUSTOMER

Home

PROFILE

RESERVATION FORM

RESERVATION INFO

FOOD

BEVERAGE

YOUR ORDER

siti.alshah@gmail.com

RESERVATION INFO

LATEST RESERVATION

RESERVATION DATE: 04-JUL-2024 00:00:00

RESERVATION ID: 60

Use a reservation ID to place a single order or multiple orders.

RESERVATION LIST

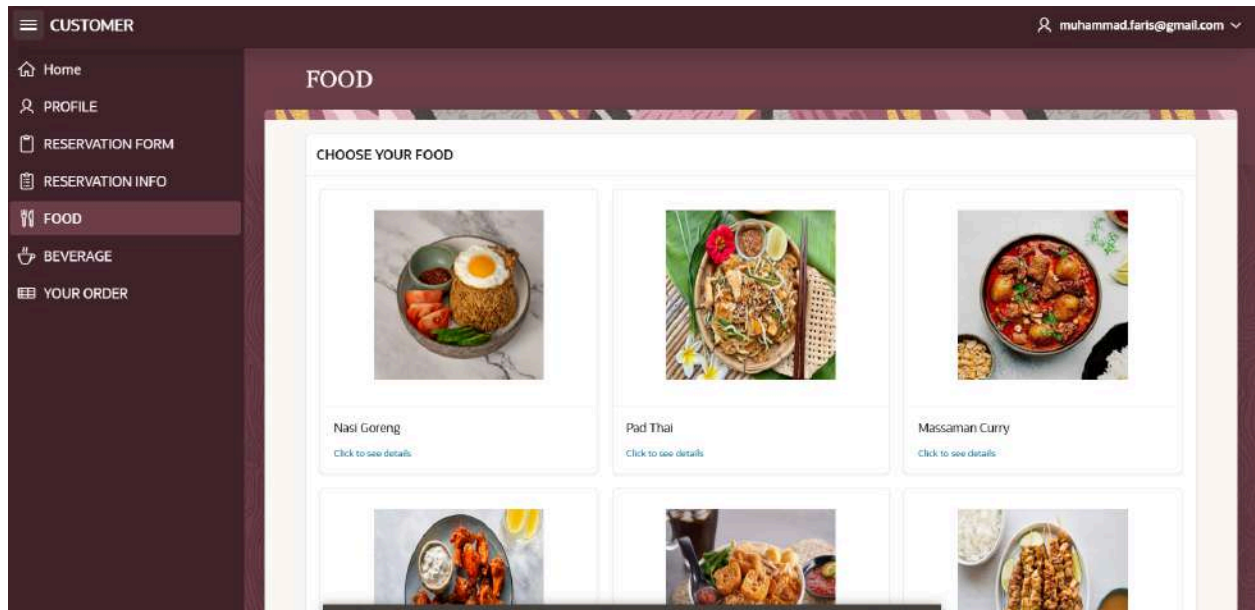
Reservation Date	Reservation ID
7/4/2024	60

1 x 1

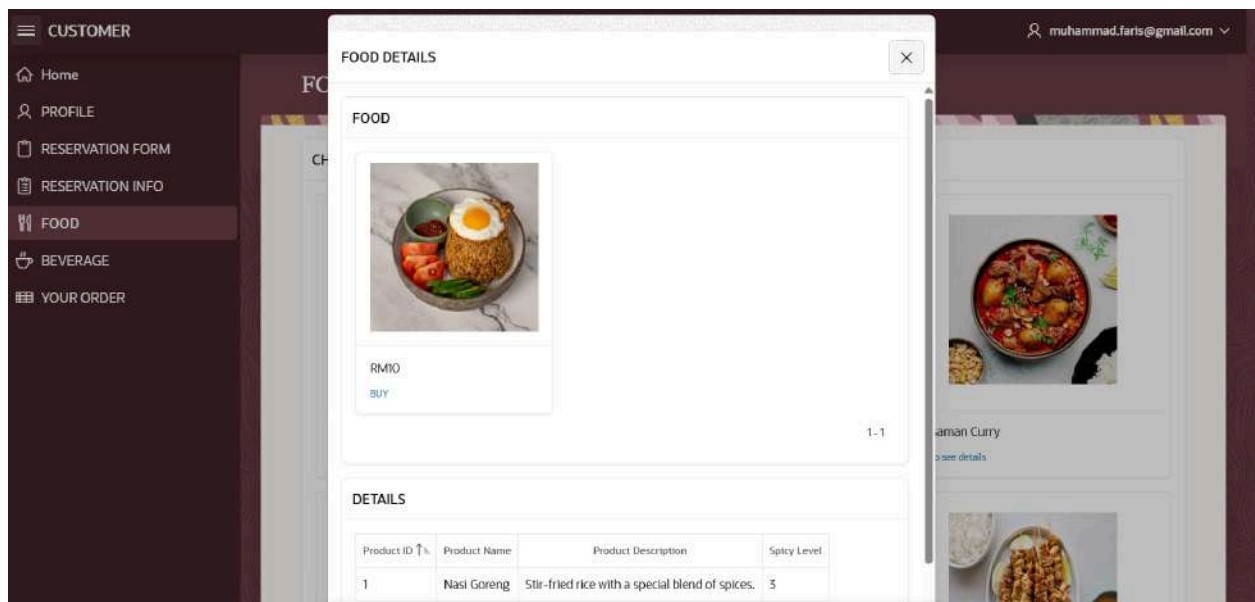
Order Food

Order Beverage

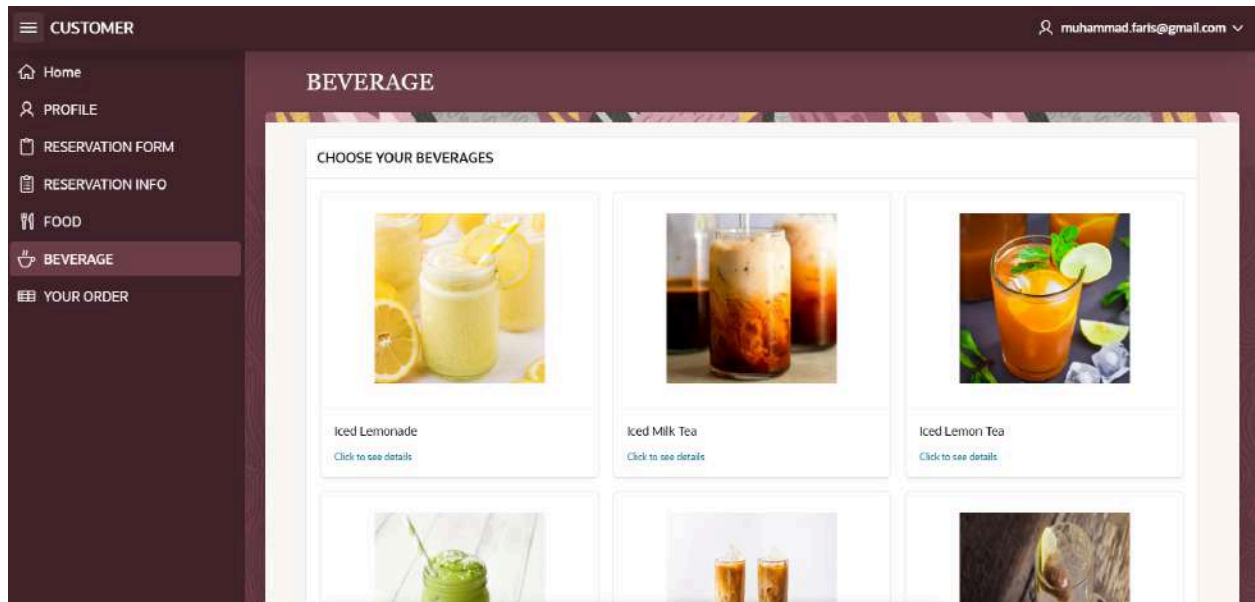
RESERVATION INFO PAGE



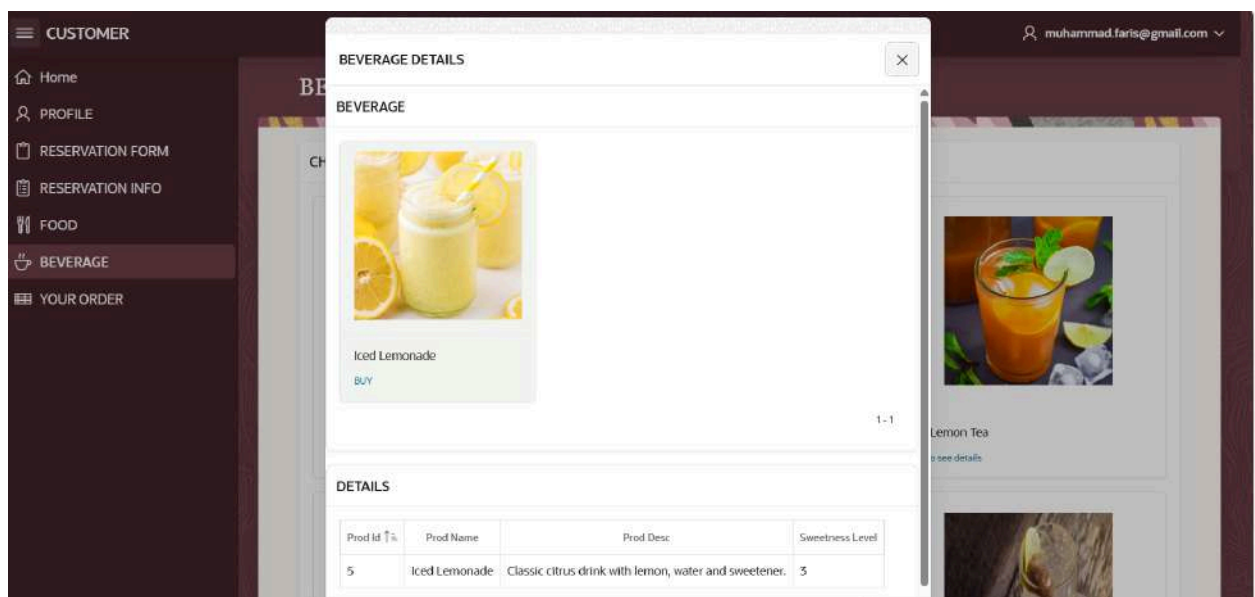
FOOD LIST PAGE



FOOD DETAILS PAGE



BEVERAGE LIST PAGE



BEVERAGE DETAILS

BEVERAGE

Iced Lemonade
[BUY](#)

1 - 1

DETAILS

Prod Id	Prod Name	Prod Desc	Sweetness Level
5	Iced Lemonade	Classic citrus drink with lemon, water and sweetener.	3

BEVERAGE DETAILS PAGE

CUSTOMER siti.aishah@gmail.com

- Home
- PROFILE
- RESERVATION FORM
- RESERVATION INFO
- FOOD**
- BEVERAGE
- YOUR ORDER

LIST FORM

PLACE YOUR ORDER

Product Name
Nasi Goreng Required

Reservation ID Required

Quantity Required

[Make a reservation](#) [Forgot Reservation ID](#) [Order](#)

LIST FORM PAGE

CUSTOMER muhammad.faris@gmail.com

- Home
- PROFILE
- RESERVATION FORM
- RESERVATION INFO
- FOOD
- BEVERAGE
- YOUR ORDER**

YOUR ORDER

Res Date	Prod Name	Prod Quantity
7/10/2024	Nasi Goreng	1
7/10/2024	Thai Chicken Satay	2

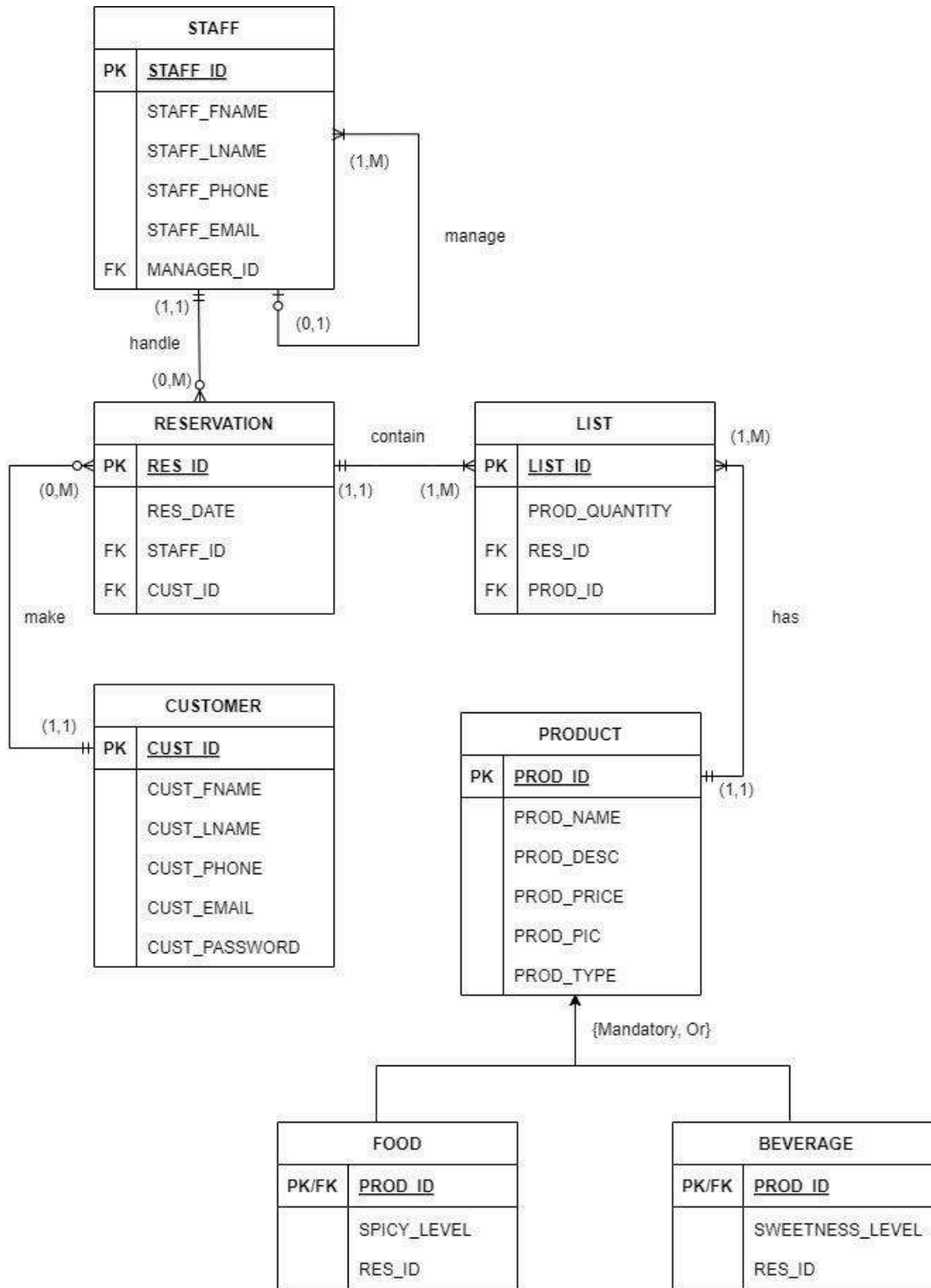
YOUR ORDER LIST PAGE

5.0 CONCLUSION

In this report, we used APEX ORACLE to create a web-based reservation and data management system for DiBo Kitchen. We identified and addressed the key requirements needed to streamline the restaurant's operations. Several factors were considered during the system's design, including a cohesive theme, user-friendly interface, and an efficient workflow. The result is a practical and straightforward solution that enhances the overall efficiency of DiBo Kitchen. The new system replaces the traditional paper-based records with a web-based system, which significantly improves the management of reservations and staff schedules. This transformation is expected to increase productivity within the restaurant's operations. The system caters to both customers and administrators, offering a seamless reservation experience for customers and efficient management tools for the staff. Moreover, the system is compatible with any operating system and web browser, ensuring accessibility and ease of use for all users. In conclusion, the implementation of this system at DiBo Kitchen reduces operational inefficiencies and enhances the dining experience, supporting the restaurant's continued growth and success.

6.0 APPENDIX

6.1 Appendix A : ERD



6.2 Appendix B : Data Dictionary

CUSTOMER

Column Name	Data Type	Nullable	Default	Primary Key
CUST_ID	NUMBER(10,0)	No	-	1
CUST_FNAME	VARCHAR2(50)	No	-	-
CUST_LNAME	VARCHAR2(50)	No	-	-
CUST_PHONE	VARCHAR2(20)	No	-	-
CUST_EMAIL	VARCHAR2(50)	No	-	-
CUST_PASSWORD	VARCHAR2(30)	No	-	-

RESERVATION

Column Name	Data Type	Nullable	Default	Primary Key
RES_ID	NUMBER(10,0)	No	-	1
RES_DATE	DATE	No	-	-
STAFF_ID	NUMBER(10,0)	Yes	-	-
CUST_ID	NUMBER(10,0)	Yes	-	-

STAFF

Column Name	Data Type	Nullable	Default	Primary Key
STAFF_ID	NUMBER(10,0)	No	-	1
STAFF_FNAME	VARCHAR2(50)	No	-	-
STAFF_LNAME	VARCHAR2(50)	No	-	-
STAFF_PHONE	VARCHAR2(11)	No	-	-
STAFF_EMAIL	VARCHAR2(50)	No	-	-
MANAGER_ID	NUMBER(10,0)	Yes	-	-

LIST

Column Name	Data Type	Nullable	Default	Primary Key
LIST_ID	NUMBER(10,0)	No	-	1
PROD_ID	NUMBER(10,0)	No	-	-
RES_ID	NUMBER(10,0)	No	-	-
PROD_QUANTITY	NUMBER(2,0)	No	-	-

PRODUCT

Column Name	Data Type	Nullable	Default	Primary Key
PROD_ID	NUMBER(10,0)	No	-	1
PROD_NAME	VARCHAR2(50)	No	-	-
PROD_DESC	VARCHAR2(100)	Yes	-	-
PROD_PRICE	NUMBER(10,0)	No	-	-
PROD_PIC	BLOB	Yes	-	-
PROD_TYPE	VARCHAR2(50)	Yes	-	-

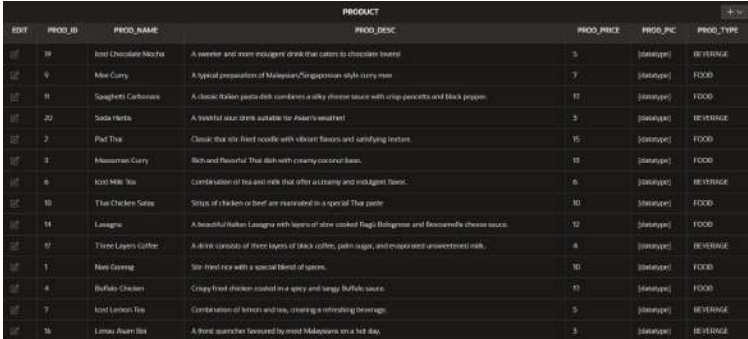
FOOD

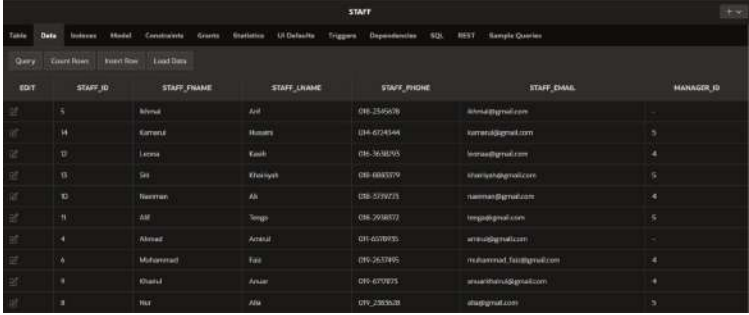
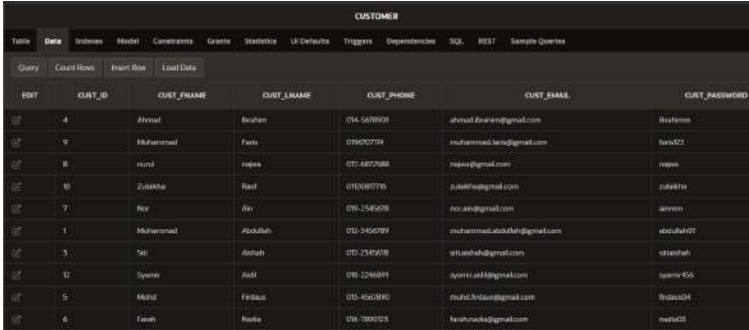
Column Name	Data Type	Nullable	Default	Primary Key
PROD_ID	NUMBER(10,0)	No	-	1
SPICY_LEVEL	NUMBER(1,0)	No	-	-

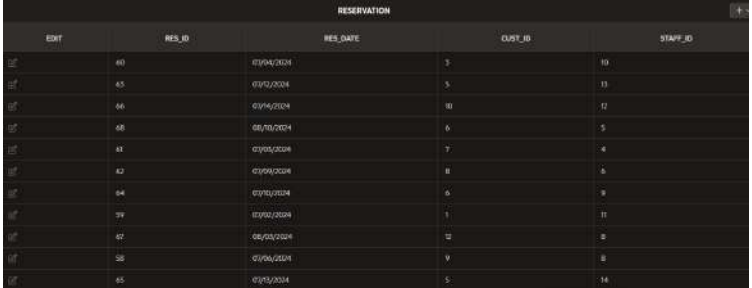
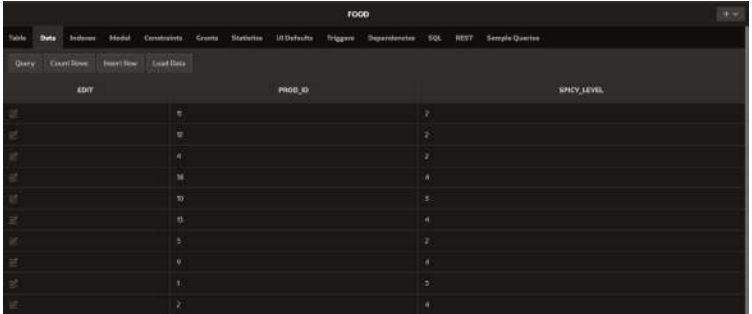
BEVERAGE

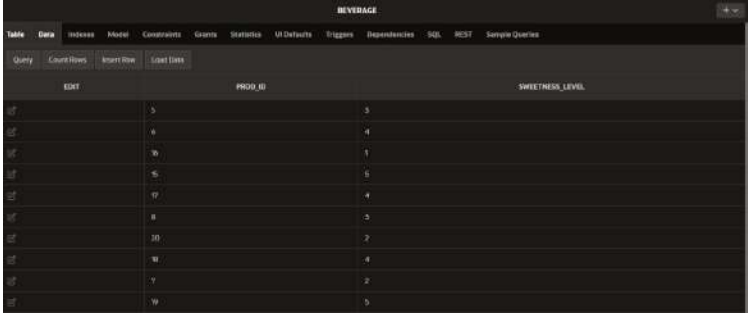
Column Name	Data Type	Nullable	Default	Primary Key
PROD_ID	NUMBER(10,0)	No	-	1
SWEETNESS_LEVEL	NUMBER(1,0)	No	-	-

6.3 Appendix C : DDL

NUM	DDL	SQL CODE
1	CREATE TABLE PRODUCT	<pre>CREATE TABLE PRODUCT(PROD_ID number(10) constraint prodid_pk primary key, PROD_NAME varchar2 (50) constraint prodname_nn not null, PROD_PRICE number (6,2) constraint prodprice_nn not null, PROD_DESC varchar2 (100) constraint proddesc_nn not null);</pre> 
2	CREATE TABLE STAFF	<pre>create table STAFF (STAFF_ID number(10) primary key, STAFF_FNAME varchar2(50) not null, STAFF_LNAME varchar2(50) not null, STAFF_PHONE varchar2(11) not null, STAFF_EMAIL varchar2(50) not null, MANAGER_ID number(10), constraint FK_MANAGER_ID foreign key (MANAGER_ID) references STAFF(STAFF_ID), constraint STAFF_PHONE_UK unique (STAFF_PHONE), constraint STAFF_EMAIL_UK unique (STAFF_EMAIL));</pre>

		
3	CREATE TABLE CUSTOMER	CREATE TABLE CUSTOMER (CUST_ID number(10) NOT NULL, CUST_FNAME varchar2(50) NOT NULL, CUST_LNAME varchar2(50) NOT NULL, CUST_PHONE varchar2(20) NOT NULL, CUST_EMAIL varchar2(50) NOT NULL, CUST_PASSWORD varchar2(30) NOT NULL, CONSTRAINT CUSTOMER_PK PRIMARY KEY (CUST_ID), CONSTRAINT CUST_PHONE_UK UNIQUE (CUST_PHONE), CONSTRAINT CUST_EMAIL_UK UNIQUE (CUST_EMAIL)); 
4	CREATE TABLE RESERVATION	CREATE TABLE RESERVATION (RES_ID number(10,0), RES_DATE date constraint RESDATE_NN not null, CUST_ID number(10,0) constraint CUSTID_NN not null, PROD_ID number(10,0) constraint PRODID_NN not null, STAFF_ID number(10,0) constraint STAFFID_NN not null, primary key (RES_ID), constraint CUST_ID_FK1 foreign key(CUST_ID)

		references CUSTOMER (CUST_ID), constraint PROD_ID_FK1 foreign key(PROD_ID) references PRODUCT (PROD_ID), constraint STAFF_ID_FK1 foreign key(STAFF_ID) references STAFF (STAFF_ID)); 
5	CREATE TABLE FOOD	CREATE TABLE FOOD (PROD_ID number(10,0), SPICY_LEVEL number(1,0), Constraint prodid_pk2 primary key (PROD_ID), Constraint spicylevel_nn check (SPICY_LEVEL is not null), Constraint prodid_fk2 foreign key (PROD_ID) REFERENCES PRODUCT); 
6	CREATE TABLE BEVERAGE	CREATE TABLE BEVERAGE (PROD_ID number(10,0), SWEETNESS_LEVEL number(1,0), Constraint prodid_pk3 primary key (PROD_ID), Constraint sweetnesslevel_nn check (SWETNESS_LEVEL is not null), Constraint prodid_fk3 foreign key (PROD_ID) REFERENCES PRODUCT);

		
7	DROP TABLE PRODUCT	DROP TABLE PRODUCT;
8	DROP TABLE STAFF	DROP TABLE STAFF;
9	DROP TABLE CUSTOMER	DROP TABLE CUSTOMER;
10	DROP TABLE RESERVATION	DROP TABLE RESERVATION;
11	DROP TABLE FOOD	DROP TABLE FOOD;
12	DROP TABLE BEVERAGE	DROP TABLE BEVERAGE;

6.4 Appendix D : DML

1. SELECT, INSERT, UPDATE, DELETE table CUSTOMER

- SELECT CUST_FNAME || ' ' || CUST_LNAME AS "Full Name"
FROM CUSTOMER
WHERE CUST_EMAIL = 'siti.aishah@gmail.com';

```
1 SELECT CUST_FNAME || ' ' || CUST_LNAME AS "Full Name"
2 FROM CUSTOMER
3 WHERE CUST_EMAIL = 'siti.aishah@gmail.com';
```

Results Explain Describe Saved SQL History

Full Name
Siti Aishah

- INSERT INTO CUSTOMER(CUST_ID, CUST_FNAME,CUST_LNAME,
CUST_PHONE,CUST_EMAIL)
VALUES (custid_seq.NEXTVAL, 'Syamir', 'Aidil', '018-2246891',
'syamir.aidil@gmail.com');

```
1 INSERT INTO CUSTOMER( CUST_ID, CUST_FNAME,CUST_LNAME, CUST_PHONE,CUST_EMAIL,CUST_PASSWORD)
2 VALUES (custid_seq.NEXTVAL, 'Syamir', 'Aidil', '018-2246891', 'syamir.aidil@gmail.com', 'syamir456');
```

Results Explain Describe Saved SQL History

1 row(s) inserted.
0.01 seconds

- UPDATE CUSTOMER
SET CUST_FNAME = 'Zulaikha',
CUST_LNAME = 'Raof',
CUST_EMAIL = 'zulaikha@gmail.com'
WHERE CUST_ID = 10;

```
1 UPDATE CUSTOMER
2 SET CUST_FNAME = 'Zulaikha',
3 CUST_LNAME = 'Raof',
4 CUST_EMAIL = 'zulaikha@gmail.com'
5 WHERE CUST_ID = 10;
```

Results Explain Describe Saved SQL History

1 row(s) updated.
0.00 seconds

- DELETE FROM CUSTOMER
WHERE CUST_ID = 2;

```
1
2 DELETE FROM CUSTOMER
3 WHERE CUST_ID = 2;
4
```

Results Explain Describe Saved SQL History

1 row(s) deleted.
0.00 seconds

2. SELECT, INSERT, UPDATE, DELETE table STAFF

- SELECT STAFF_FNAME || ' ' || STAFF_LNAME AS "Full Name", STAFF_EMAIL
FROM STAFF
WHERE MANAGER_ID = 4;

```
1 SELECT STAFF_FNAME || ' ' || STAFF_LNAME AS "Full Name", STAFF_EMAIL
2 FROM STAFF
3 WHERE MANAGER_ID = 4;
4
```

Full Name	STAFF_EMAIL
Leona Kasih	leonaa@gmail.com
Noeman Ali	noeman@gmail.com
Muhammad Faiz	muhammad_faiz@gmail.com
Khairul Anuar	anuar_khairul@gmail.com

4 rows returned in 0.01 seconds [Download](#)

- INSERT INTO STAFF(STAFF_ID, STAFF_FNAME, STAFF_LNAME,
STAFF_PHONE, STAFF_EMAIL, MANAGER_ID)
VALUE(STAFFID_SEQ.NEXTVAL, 'Alif', 'Teega', '018-2938372',
'teega@gmail.com', 5);

```
1 INSERT INTO STAFF (STAFF_ID, STAFF_FNAME, STAFF_LNAME, STAFF_PHONE, STAFF_EMAIL, MANAGER_ID)
2 VALUES (STAFFID_SEQ.NEXTVAL, 'Kamarul', 'Husaini', '014-6724344', 'kamarul@gmail.com', 5);
3
```

1 row(s) inserted.

0.00 seconds

- UPDATE STAFF
SET STAFF_PHONE = '019-6717873'
WHERE STAFF_ID = 9;

```
1 SET STAFF_PHONE = '019-6717873'
2 WHERE STAFF_ID = 9;
3
4
5
```

1 row(s) updated.

0.07 seconds

- DELETE FROM STAFF
WHERE STAFF_ID = 7;

```
1 DELETE FROM STAFF
2 WHERE STAFF_ID = 7;
3
```

1 row(s) deleted.

0.01 seconds

3. SELECT, INSERT, UPDATE, DELETE table FOOD

- SELECT SPICY_LEVEL
FROM FOOD
WHERE PROD_ID = 9;

```
1 SELECT SPICY_LEVEL
2 FROM FOOD
3 WHERE PROD_ID = 9;
4
```

SPICY_LEVEL
4

1 rows returned in 0.01 seconds [Download](#)

- UPDATE FOOD
SET SPICY_LEVEL=3
WHERE PROD_ID=1;

```
1 UPDATE FOOD
2 SET SPICY_LEVEL=3
3 WHERE PROD_ID=1;
4
```

1 row(s) updated.

0.00 seconds

4. SELECT, INSERT, UPDATE, DELETE table BEVERAGE

- SELECT SWEETNESS_LEVEL
FROM BEVERAGE
WHERE PROD_ID = 19;

```
1 SELECT SWEETNESS_LEVEL
2 FROM BEVERAGE
3 WHERE PROD_ID = 19;
4
```

SWEETNESS_LEVEL
5

1 rows returned in 0.00 seconds [Download](#)

- UPDATE BEVERAGE
SET SWEETNESS_LEVEL=1
WHERE PROD_ID=16;

```
1 UPDATE BEVERAGE
2 SET SWEETNESS_LEVEL=1
3 WHERE PROD_ID=16;
4
```

1 row(s) updated.

0.01 seconds

5. SELECT, INSERT, UPDATE, DELETE table PRODUCT

- SELECT PROD_NAME, PROD_DESC, PROD_PRICE
FROM PRODUCT
WHERE PROD_ID=14;

```
1 SELECT PROD_NAME, PROD_DESC, PROD_PRICE
2 FROM PRODUCT
3 WHERE PROD_ID=14;
4
```

PROD_NAME	PROD_DESC	PROD_PRICE
Lasagna	A beautiful Italian Lasagna with layers of slow cooked Ragù Bolognese and Besciamella cheese sauce.	12

1 rows returned in 0.01 seconds

- INSERT INTO PRODUCT(PROD_ID, PROD_NAME, PROD_DESC,
PROD_PRICE, PROD_TYPE)
VALUE(PRODID_SEQ.NEXTVAL, 'Plain Water', 'A freshful drink suitable for
Asian's weather!', 0, BEVERAGE);

```
1 INSERT INTO PRODUCT (PROD_ID, PROD_NAME, PROD_DESC, PROD_PRICE, PROD_TYPE)
2 VALUES (PRODID_SEQ.NEXTVAL, 'Plain Water', 'A freshful drink suitable for Asian weather!', 0, 'BEVERAGE');
3
```

1 row(s) inserted.

0.00 seconds

- UPDATE PRODUCT
SET PROD_PRICE=12
WHERE PROD_ID=14;

```
1 UPDATE PRODUCT
2 SET PROD_PRICE=12
3 WHERE PROD_ID=14;
4
```

1 row(s) updated.

0.01 seconds

- DELETE FROM BEVERAGE
WHERE PROD_ID=9;

```
1 DELETE FROM BEVERAGE
2 WHERE PROD_ID=9;
3
```

0 row(s) deleted.

0.00 seconds

6. SELECT, INSERT, UPDATE, DELETE table RESERVATION

- SELECT RES_DATE
FROM RESERVATION
WHERE RES_ID=35

```
1 SELECT RES_DATE
2 FROM RESERVATION
3 WHERE RES_ID=35
4
5
```

RES_DATE
07/02/2024

1 rows returned in 0.01 seconds [Download](#)

- INSERT INTO RESERVATION (RES_ID, RES_DATE)
VALUES (RESERVATION_SEQ.NEXTVAL, TO_DATE('10/07/2024',
'DD/MM/YYYY'));

```
1 INSERT INTO RESERVATION (RES_ID, RES_DATE)
2 VALUES (RESERVATION_SEQ.NEXTVAL, TO_DATE('10/07/2024', 'DD/MM/YYYY'));
3
```

1 row(s) inserted.
0.00 seconds

- UPDATE RESERVATION
SET RES_DATE='07/01/2024'
WHERE RES_ID=54;

```
1 UPDATE RESERVATION
2 SET RES_DATE='07/01/2024'
3 WHERE RES_ID=54;
4
```

1 row(s) updated.
0.01 seconds

- DELETE FROM RESERVATION
WHERE RES_ID=15;

```
1 DELETE FROM RESERVATION
2 WHERE RES_ID=15;
3
```

1 row(s) deleted.
0.01 seconds

7. SELECT, INSERT, UPDATE, DELETE table LIST

- SELECT RES_ID, PROD_QUANTITY
FROM LIST
WHERE LIST_ID=2;

```
1 SELECT RES_ID, PROD_QUANTITY
2 FROM LIST
3 WHERE LIST_ID=2;
```

RES_ID	PROD_QUANTITY
63	1

1 rows returned in 0.00 seconds [Download](#)

- INSERT INTO LIST (LIST_ID, RES_ID, PROD_QUANTITY)
VALUES (ORDERID_SEQ.NEXTVAL, 58, 2);

```
1 INSERT INTO LIST (LIST_ID, RES_ID, PROD_QUANTITY)
2 VALUES (ORDERID_SEQ.NEXTVAL, 58, 2);
3
```

1 row(s) inserted.

0.00 seconds

- UPDATE LIST
SET PROD_QUANTITY=2
WHERE LIST_ID=9;

```
1 UPDATE LIST
2 SET PROD_QUANTITY=2
3 WHERE LIST_ID=9;
4
```

1 row(s) updated.

0.00 seconds

- DELETE FROM LIST
WHERE LIST_ID=58;

```
1 DELETE FROM LIST
2 WHERE LIST_ID=58;
3
```

1 row(s) deleted.

0.00 seconds

7.0 REFERENCES

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